

Solubility Silver Halides in Aqueous Medium

Compound	Solubility Product
AgF	205
AgCl	1.8×10^{-10}
AgBr	5.2×10^{-13}
AgI	8.3×10^{-17}

Acid–Base Theory

The Arrhenius Theory (1880-1890)



An acid is any hydrogen-containing compound that provides hydrogen ion in aqueous medium. Similarly, according to the theory, a base is any compound that gives hydroxyl ion in aqueous medium.



Demerits:

(A) **Acid and base properties are dependent on the presence of water (aqueous). With the help of this theory, these properties were not explainable in non-aqueous medium or in gaseous state of a substance.**

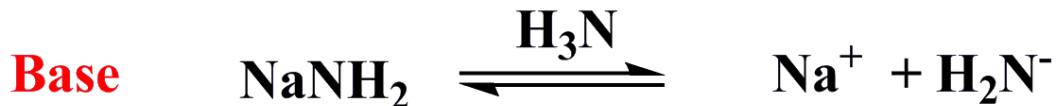
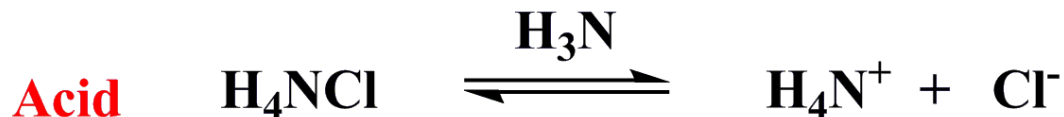
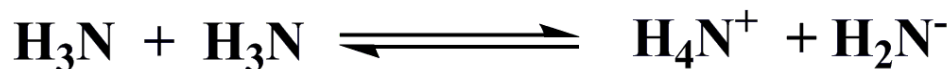
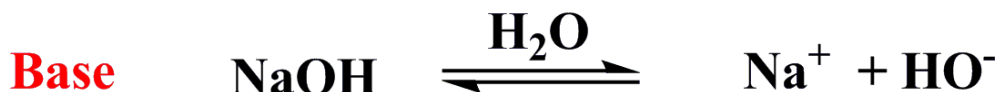
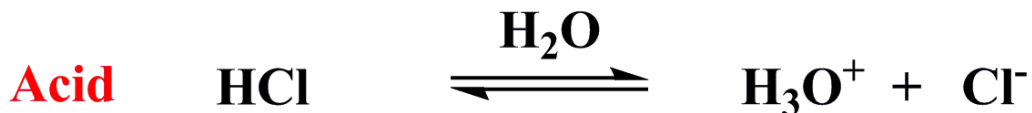
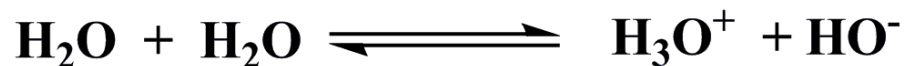
(B) The existence of bare H^+ in aqueous medium was questioned. The size of proton is extremely small and therefore, it was proposed that it combined with water molecule and exists as H_3O^+ (hydronium) ion.

Franklin Theory (1905)

This theory concerns the acidic and basic properties of a substance in non-aqueous medium, mainly, in ammonium.

According to this theory,

An acid is a solute that gives the cationic character of the solvent, and a base is a solute that provides anionic character of the solvent.

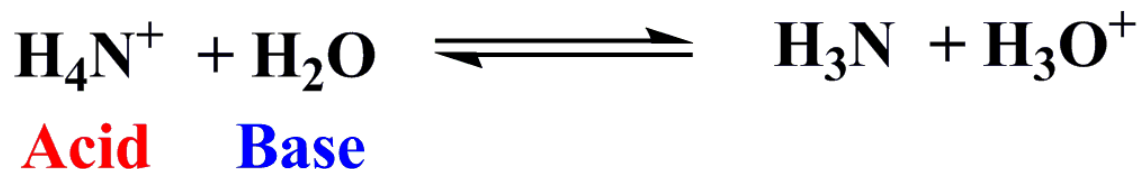
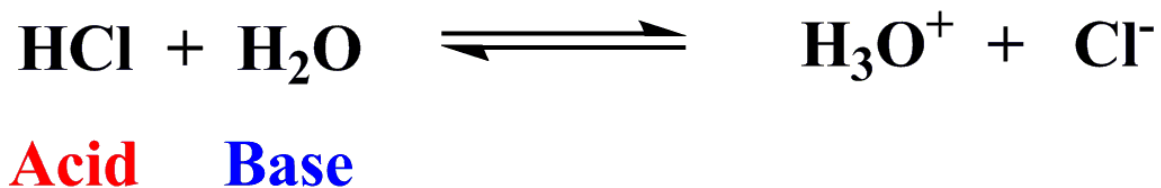


The Protonic Theory (Brønsted-Lowry concept, 1923)

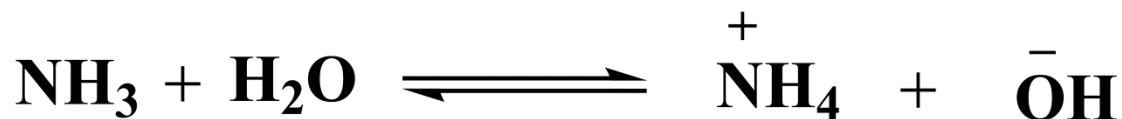


Any species that tends to **give up a proton is an acid**, and any species that tends **to accept a proton is a base**.

Examples:



Base **Acid**



Conjugate Acid-Base pair

Two chemical species, which are inter-convertible to each other by means of a proton, are known as conjugate acid-base pairs.



Demerit:

In this theory, the acid-base properties are determined **solely by proton exchange parameter**. Therefore, some other substances that do not contain any proton but have inherent acidic or basic properties, *e.g.* BF_3 , I_2 , AlCl_3 , etc cannot be explained with the help of the theory.



The Lewis acid-base definition (1923)

Lewis proposed a definition of acid-base behavior of a substance in terms of electron-pair donation and acceptance. According to him, **an acid is any species that accepts an electron pair forming a covalent bond, while; a base is any species that can donate a pair of electrons to build a covalent bond.** In short, Lewis acid is electron pair acceptor and Lewis base is electron pair donor.

Examples (Lewis acid):

(i) Molecules containing an atom processing incomplete valence shell.

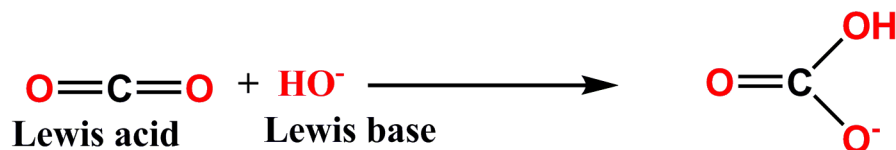
Example: BF_3 , AlCl_3 , PCl_3 , SO_3 , etc.

(ii) Cations with a stable, low-lying empty or partially filled orbitals. Transition metal ions are good examples for this category. Coordination of these ions with electron donating ligands are then must be consider as Lewis acid-base reactions.

Example: $\text{Cu}^+ + n\text{NH}_3 \longrightarrow [\text{Cu}(\text{NH}_3)_n]^+$

(iii) Compounds with hetero atomic double bonds.

Example:



Examples (Lewis base):

(i) Molecules having one or more unshared pair of electrons.

The basic character decreases as the number of unshared electrons holds by the donor atom increases, *e.g.* decreasing basicity order, **NH₃ (1 lone pair of electron), H₂O (2 lone pair of electron), HF (3 lone pair of electron).**

(ii) Negatively charged ions.

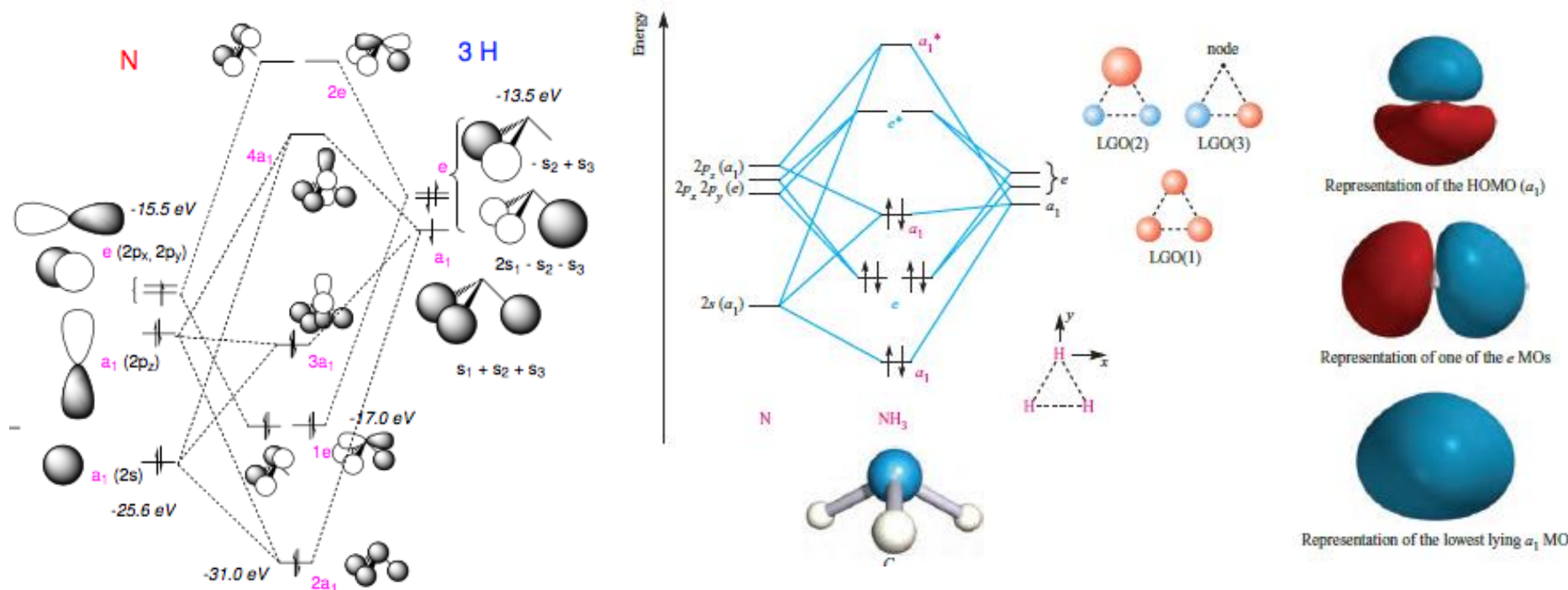
Example: F⁻, Cl⁻, I⁻, etc

(iii) Compounds containing homo atomic double bonds.

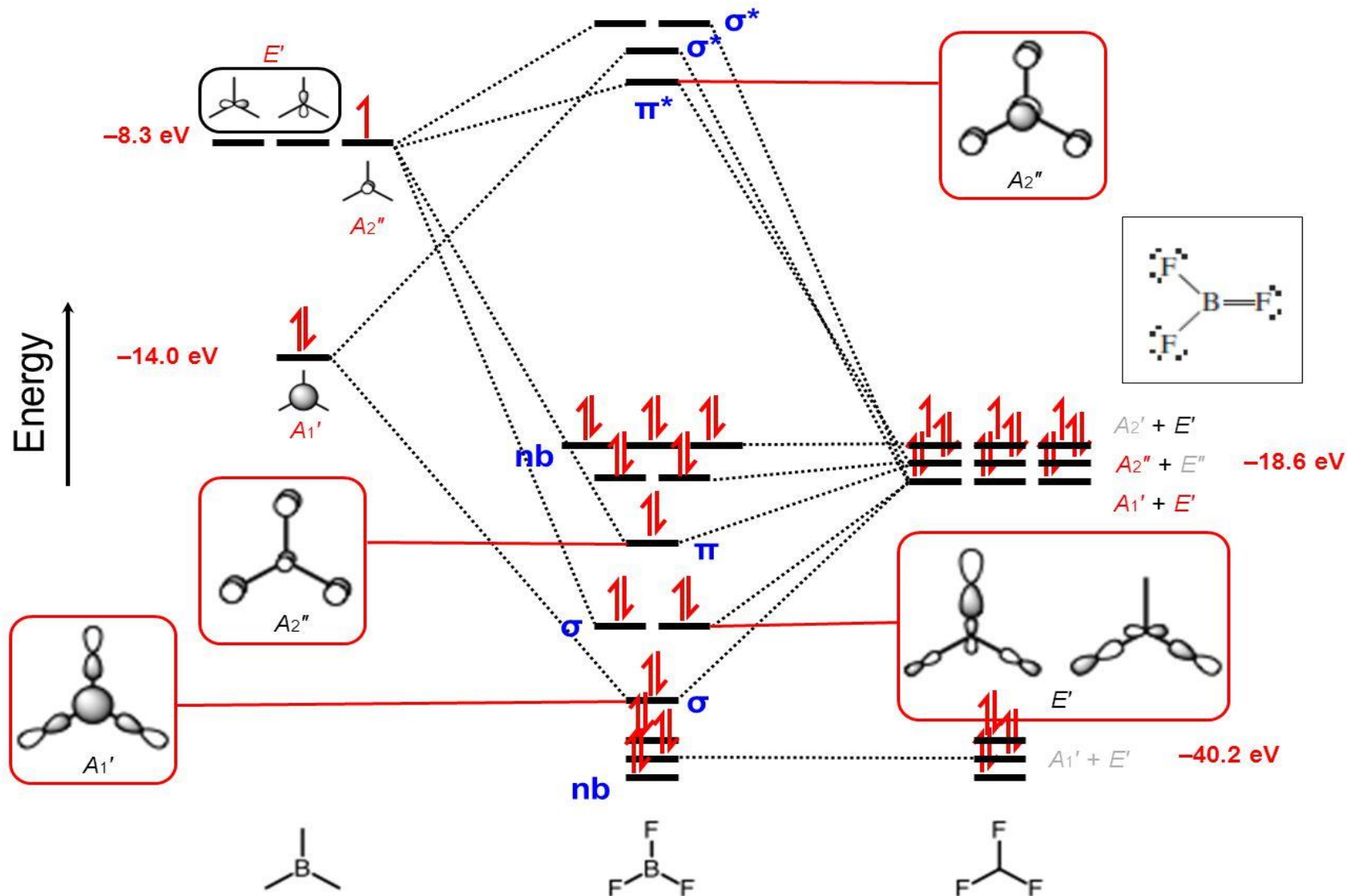
Example:

Ethylene, benzene, etc forms bonds with metals, like, Pt, Rh, Ag, etc, and forms stable acid-base complexes.

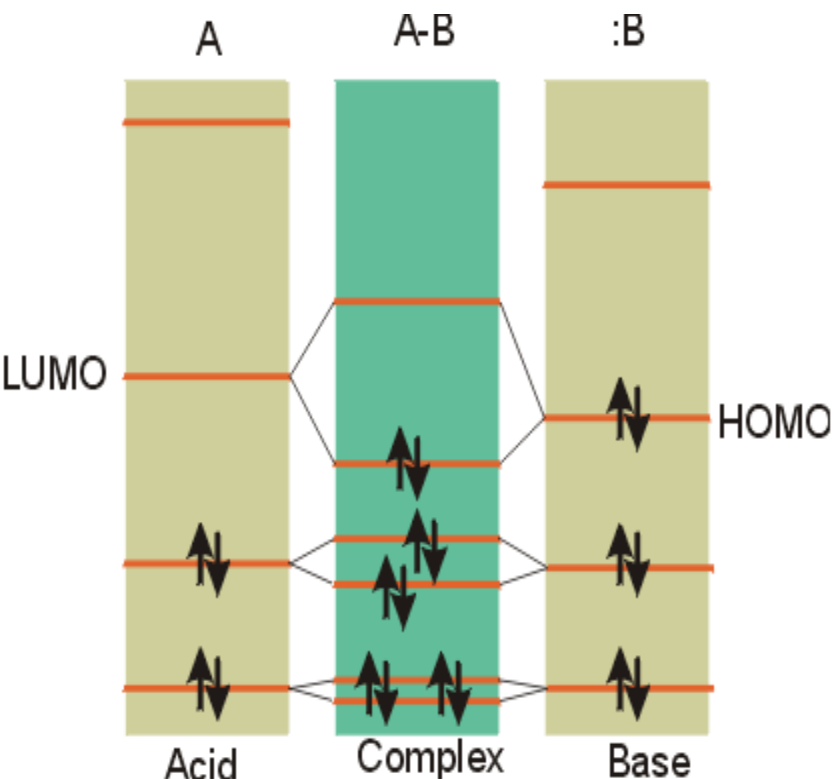
MOs of NH₃



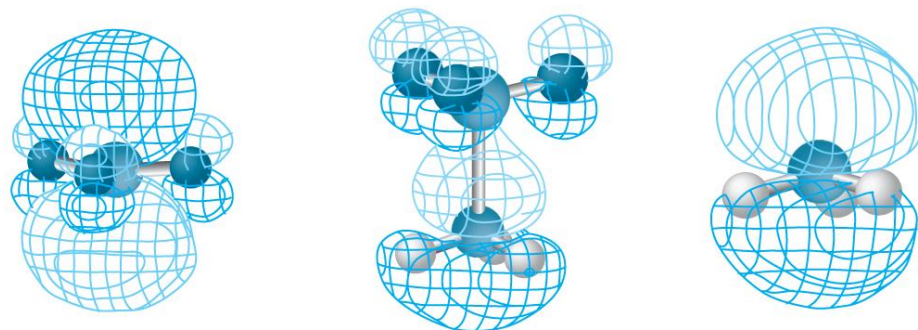
Boron Trifluoride



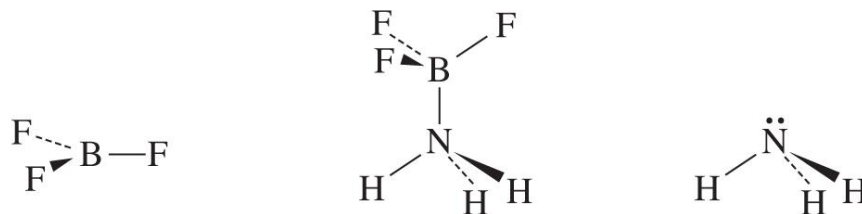
Lewis Acids and Bases: Adduct Formation



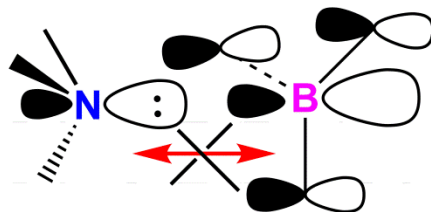
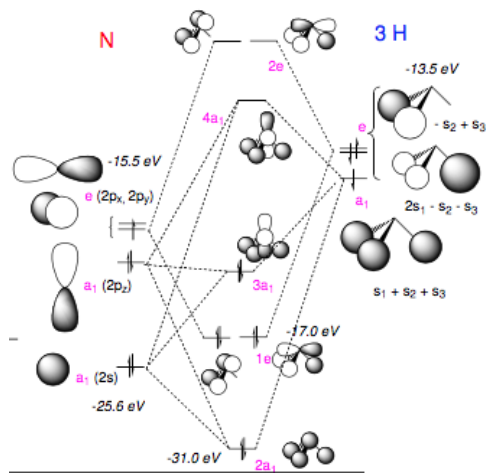
The LUMO of the acid and the HOMO of the base to form the adduct



Interaction between the LUMO of Acid and the HOMO of Base. If there is a net lowering of energy, the adduct is stable.



BF₃ and NH₃

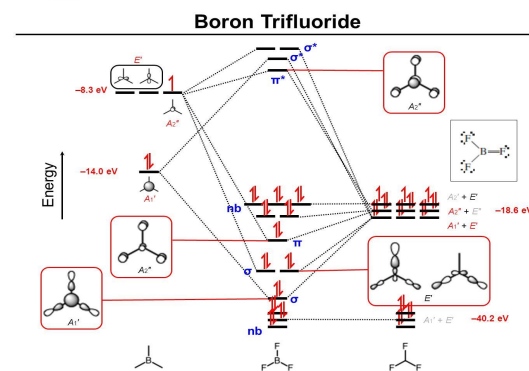
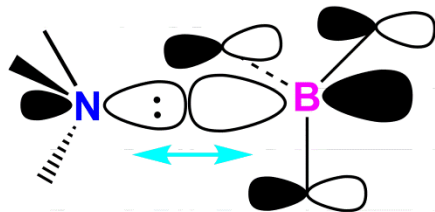
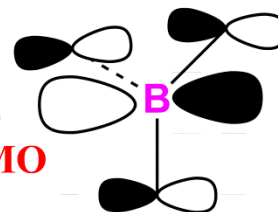


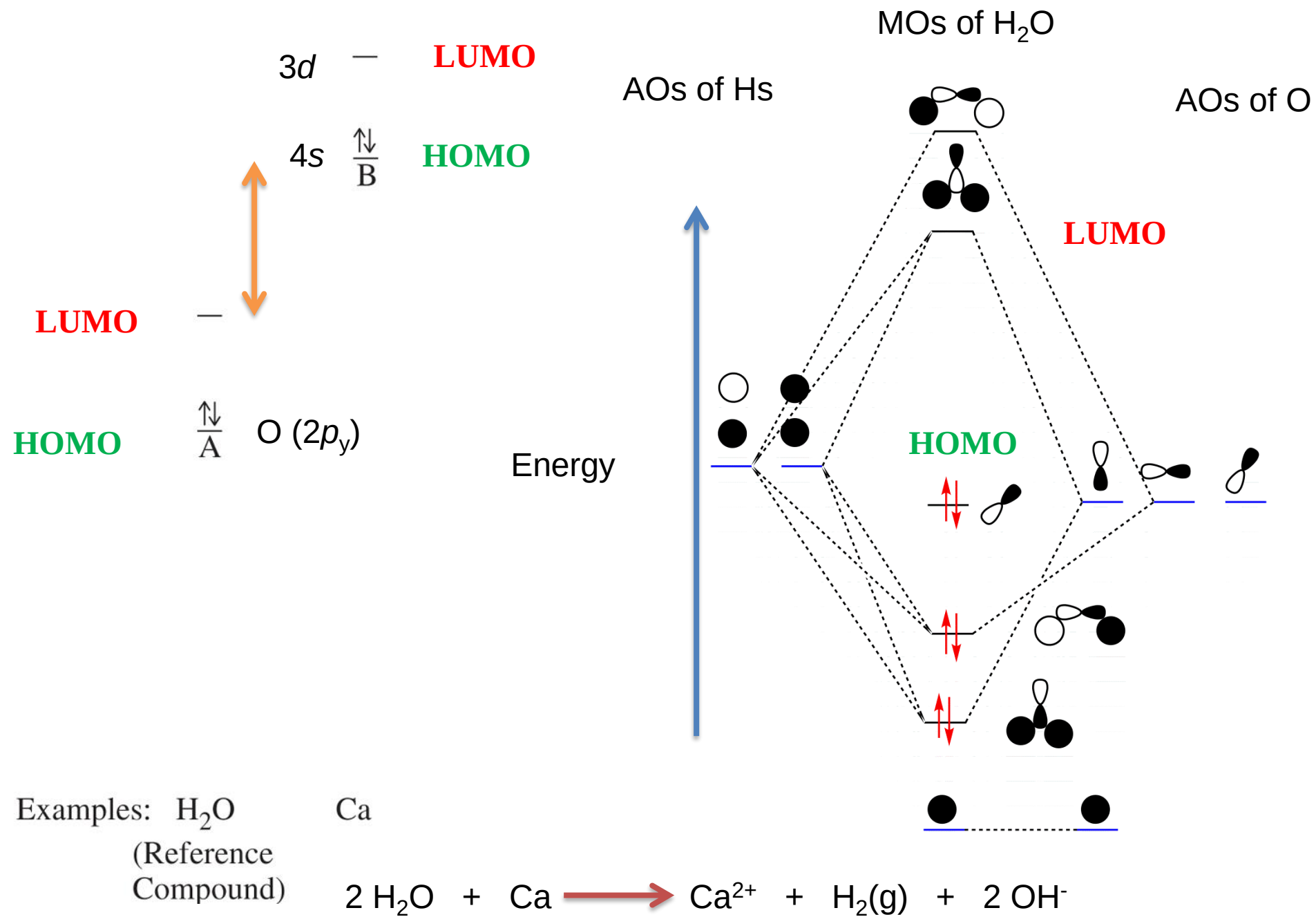
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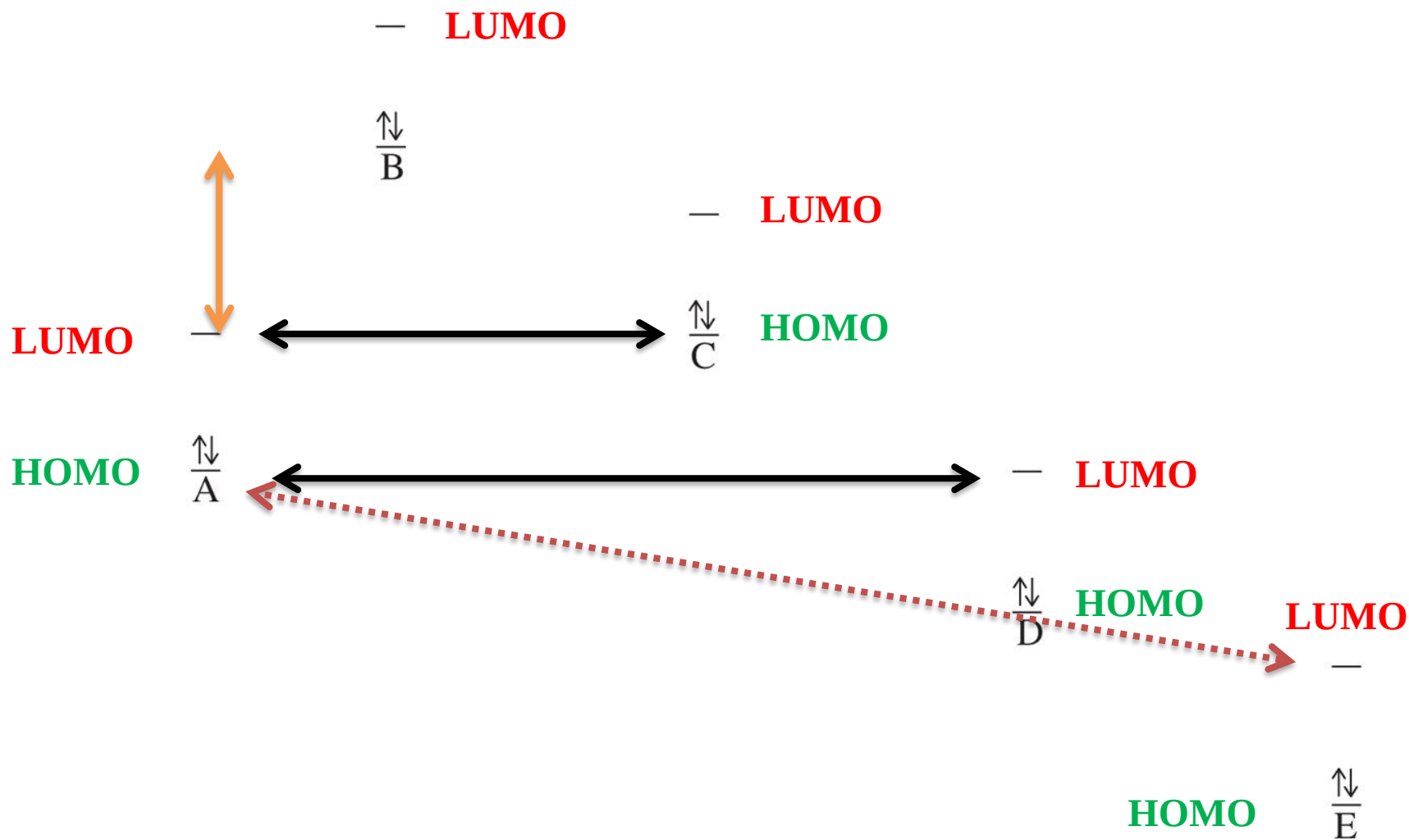
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LUMO







Examples: H₂O
(Reference
Compound)

Ca

Cl⁻

Mg²⁺

F₂



» Ca HOMO >> H₂O LUMO

» Not matched well for acid-base adduct to form

» Electron transfer (oxidation of Ca by H₂O) is the predicted reaction



» Correct energy match for acid-base adduct to form and be stable

» Water is the acid; its LUMO is used along with the base (Cl⁻) HOMO



» Correct energy match for acid-base adduct to form and be stable

» Water is the base; its HOMO is used along with the LUMO of acid Mg²⁺



» Water HOMO >> F₂ LUMO

» Not matched well for an acid-base adduct to form

» Electron transfer (reduction of F₂ by H₂O) is the predicted reaction

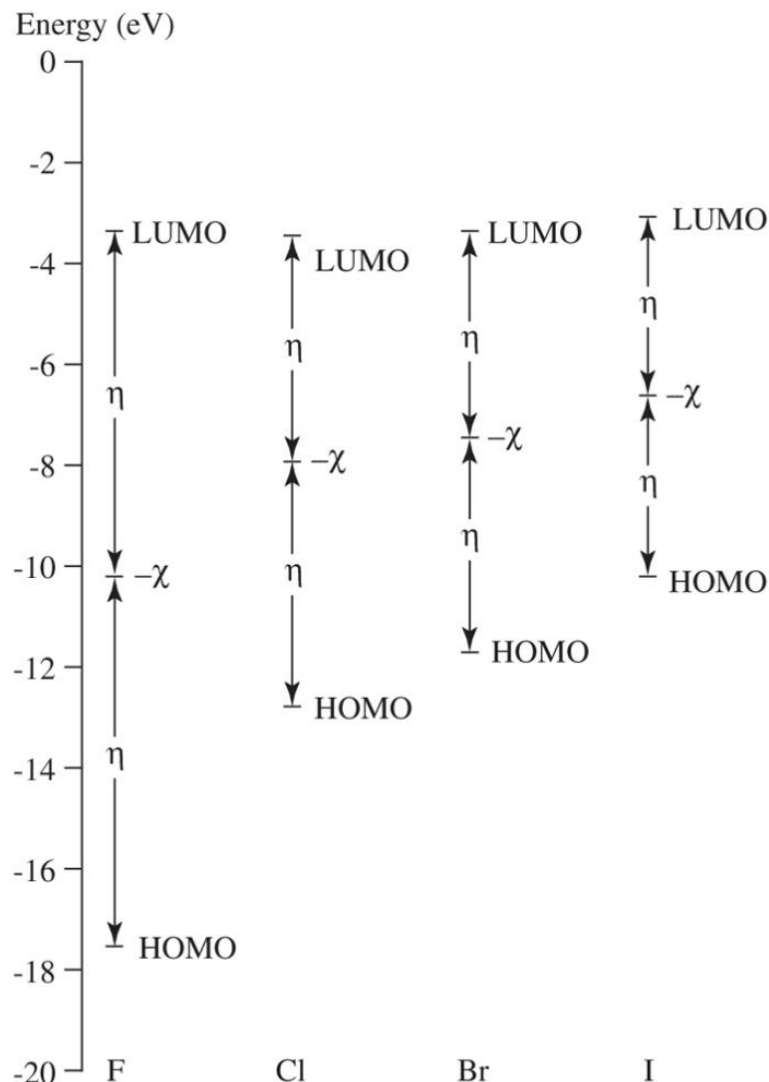
Strength of Lewis Acids and Bases

Hard (H) and Soft (S) Acid (A) Base (B) Concept [HSAB, 1963]

$$\eta = \frac{I - A}{2}$$

(I = Ionization energy and
A = Electron Affinity)

Theory: *Hard acids react preferentially with hard bases, and soft acids react preferentially with soft bases.*



Hard Acids

H^+ , Li^+ , Na^+ , K^+

Be^{2+} , Mg^{2+} , Ca^{2+}

BF_3 , BCl_3 , B(OR)_3

Al^{3+} , $\text{Al(CH}_3)_3$, AlCl_3 ,

Cr^{3+} , Mn^{2+} , Fe^{3+} , Co^{3+}

AlH_3

SO_3

Borderline

BBr_3 , $\text{B(CH}_3)_3$

Fe^{2+} , Co^{2+} , Ni^{2+}

Cu^{2+} , Zn^{2+} , Rh^{3+}

Ir^{3+} , Ru^{3+} , Os^{2+}

SO_2

Soft Acids

BH_3 , Tl^+ , $\text{Tl(CH}_3)_3$

Cu^+ , Ag^+ , Au^+ ,

Cd^{2+} , Hg_2^{2+} ,

Hg^{2+} , Pd^{2+} , Pt^{2+} ,

Pt^{4+}

Hard acids	Soft acids
(I) Small in size.	(I) Large in size.
(II) High positive oxidation state.	(II) Zero or low positive oxidation state.
(III) Absence of any outer electrons that are easily excited to higher states.	(III) Several easily excitable valence electrons.

Hard Bases

F^- , Cl^-

H_2O , OH^- , O^{2-}

ROH , RO^- , R_2O , $CH_3CO_2^-$

NO_3^- , ClO_4^-

CO_3^{2-} , SO_4^{2-} , PO_4^{3-}

NH_3 , RNH_2

Borderline

Br^-

$\underline{NO}_2^-$, N_3^- , N_2

SO_3^{2-}

$C_6H_5NH_2$, C_5H_5N

Soft Bases

H^- , I^-

H_2S , HS^- , S^{2-}

RSH , RS^- , R_2S

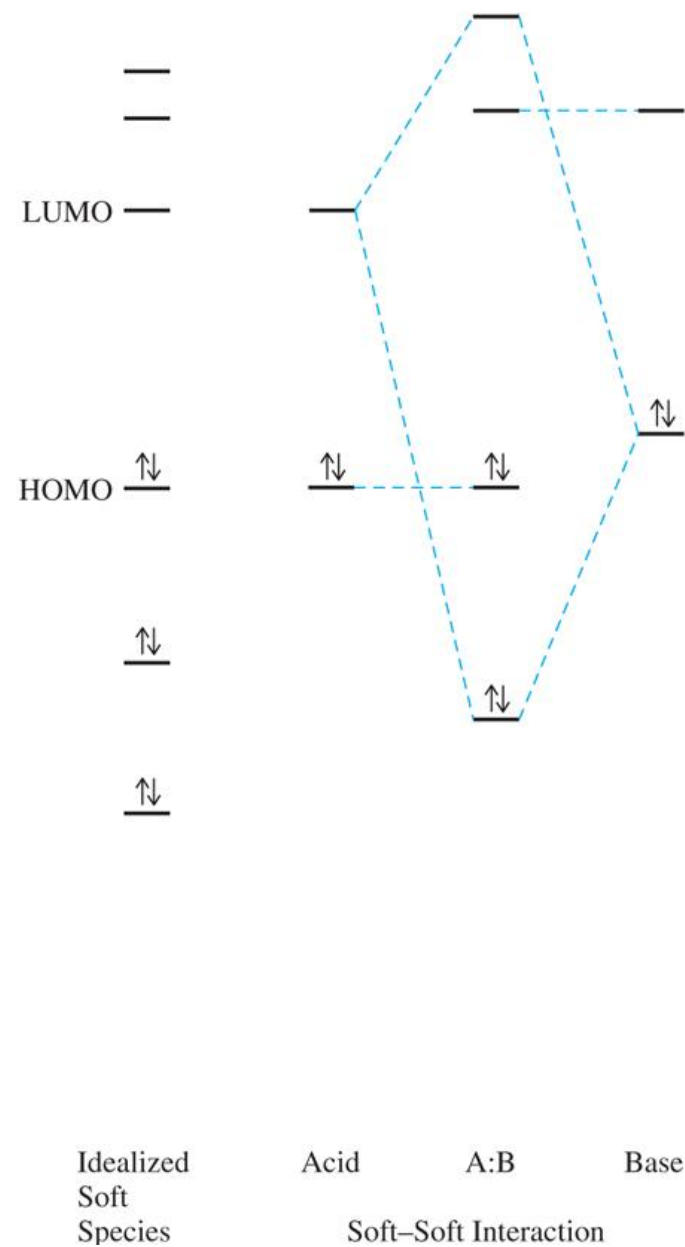
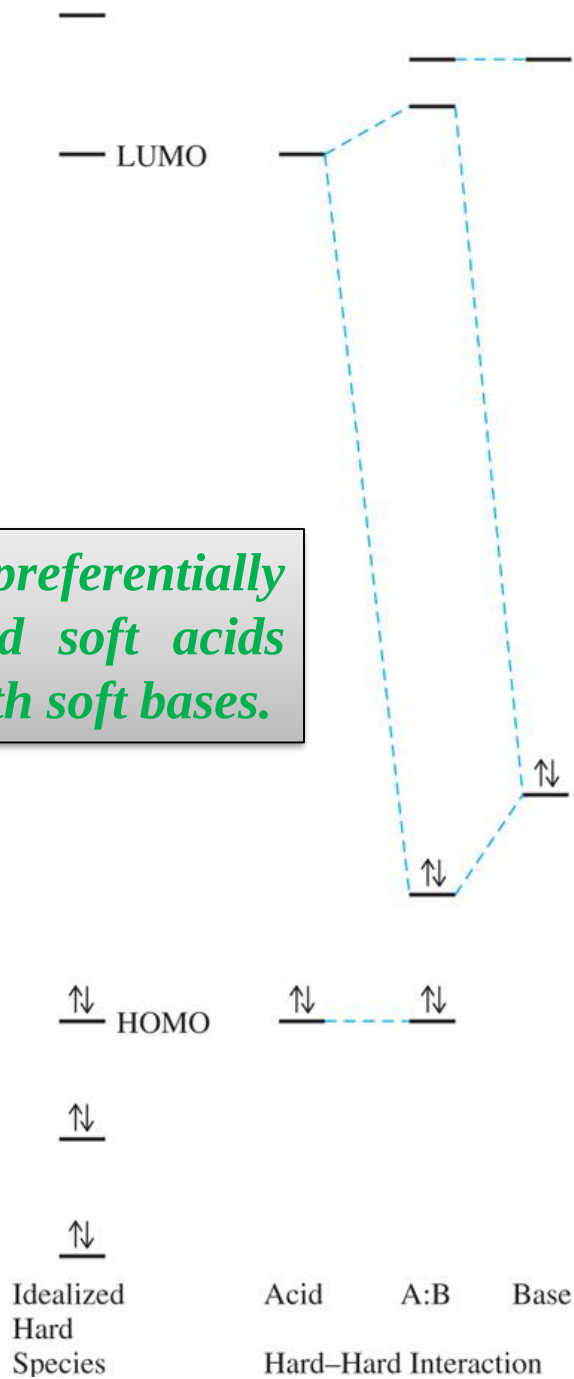
$\underline{SCN}^-$, $\underline{CN}^-$, RNC , CO

$S_2O_3^{2-}$

R_3P , C_6H_6

Hard bases	Soft bases
(I) High electronegativity.	(I) Low electronegativity.
(II) Low polarisability.	(II) High polarisability.
(III) Presence of filled orbitals; empty orbitals may exist at high energy level.	(III) Presence of partially filled orbitals; empty orbitals are low-lying.

Hard acids react preferentially with hard bases, and soft acids react preferentially with soft bases.



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AgF	205
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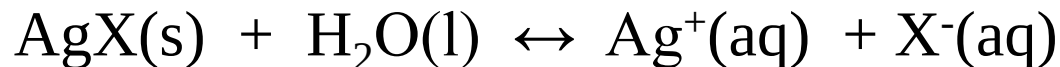


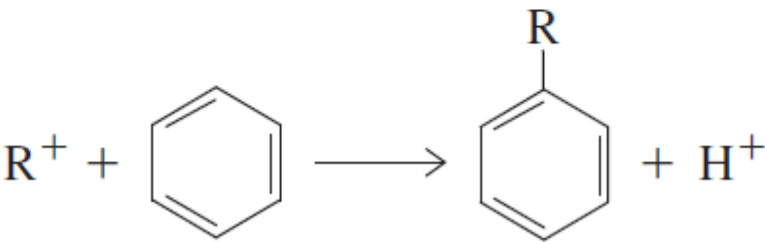
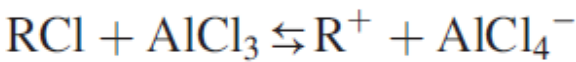
TABLE 6.3 Equilibrium Constants for Reactions of Mercury Complexes²⁵

Reaction	<i>K</i>
1. $[\text{CH}_3\text{Hg}(\text{H}_2\text{O})]^+ + \text{HF} \rightleftharpoons \text{CH}_3\text{HgF} + \text{H}_3\text{O}^+$	4.5×10^{-2}
2. $[\text{CH}_3\text{Hg}(\text{H}_2\text{O})]^+ + \text{HCl} \rightleftharpoons \text{CH}_3\text{HgCl} + \text{H}_3\text{O}^+$	1.8×10^{12}
3. $[\text{CH}_3\text{Hg}(\text{H}_2\text{O})]^+ + \text{HBr} \rightleftharpoons \text{CH}_3\text{HgBr} + \text{H}_3\text{O}^+$	4.2×10^{15}
4. $[\text{CH}_3\text{Hg}(\text{H}_2\text{O})]^+ + \text{HI} \rightleftharpoons \text{CH}_3\text{HgI} + \text{H}_3\text{O}^+$	1×10^{18}
5. $[\text{CH}_3\text{Hg}(\text{H}_2\text{O})]^+ + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{HgOH} + \text{H}_3\text{O}^+$	5×10^{-7}
6. $[\text{CH}_3\text{Hg}(\text{H}_2\text{O})]^+ + \text{SH}^- \rightleftharpoons [\text{CH}_3\text{HgS}]^- + \text{H}_3\text{O}^+$	1×10^7
7. $[\text{CH}_3\text{Hg}(\text{H}_2\text{O})]^+ + \text{HSCN} \rightleftharpoons \text{CH}_3\text{HgSCN} + \text{H}_3\text{O}^+$	5×10^6

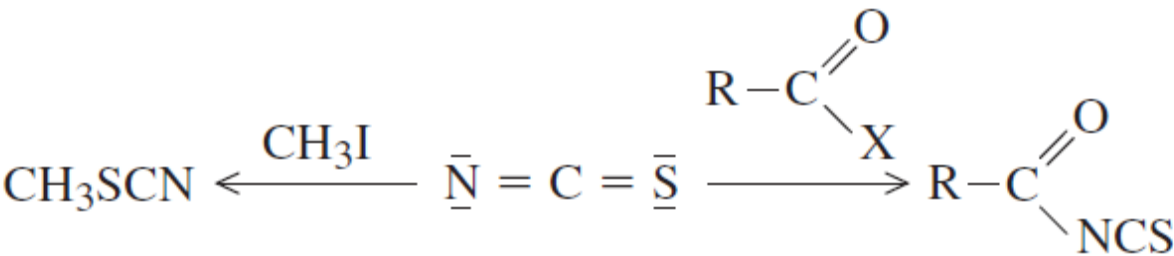
Qualitative Analysis Separation

	Group 1	Group 2	Group 3	Group 4	Group 5
HSAB acids	Soft	Borderline and soft	Borderline	Hard	Hard
Reagent	HCl	H ₂ S (acidic)	H ₂ S (basic)	(NH ₄) ₂ CO ₃	Soluble
Precipitates	AgCl	HgS	MnS	CaCO ₃	Na ⁺
	PbCl ₂	CdS	FeS	SrCO ₃	K ⁺
	Hg ₂ Cl ₂	CuS	CoS	BaCO ₃	NH ₄ ⁺
		SnS	NiS		
		As ₂ S ₃	ZnS		
		Sb ₂ S ₃	Al(OH) ₃		
		Bi ₂ S ₃	Cr(OH) ₃		

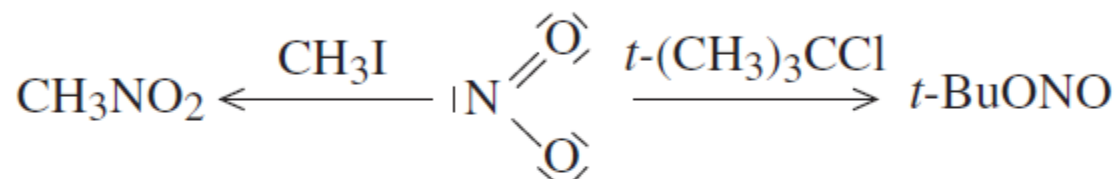
The function of the acid catalyst is to generate a *positive* attacking species. In the *Friedel-Crafts reaction*, AlCl_3 functions as an acid catalyst by increasing the concentration of the carbocation, R^+ , which is the attacking species as shown in the following equations:



Reactive site preference. We have already used the HSIP principle as it applies to linkage isomers in metal complexes. This application to bonding site preference can also be used to predict the products of some reactions. For example, the reactions of organic compounds also obey the principle when reacting with nucleophiles such as SCN^- or NO_2^- . Consider the reactions illustrated in Eq. (5.41):



In this case, the acyl group in the acidic species RCOX is a hard acid and reacts with the nitrogen end of SCN^- to form an acyl isothiocyanate. The soft methyl group in methyl iodide bonds to the S atom and forms methyl thiocyanate. Consider the following reactions of NO_2^- :



Here, the $t\text{-(CH}_3)_3\text{C}^+$ carbocation is a hard Lewis acid, so the product is determined by its interaction with the oxygen (harder) electron donor in NO_2^- . In the reaction with CH_3I , the product is nitromethane, showing the softer character of the methyl group.

Another reaction that illustrates the HSIP is the reaction of PCl_3 with AsF_3 :

